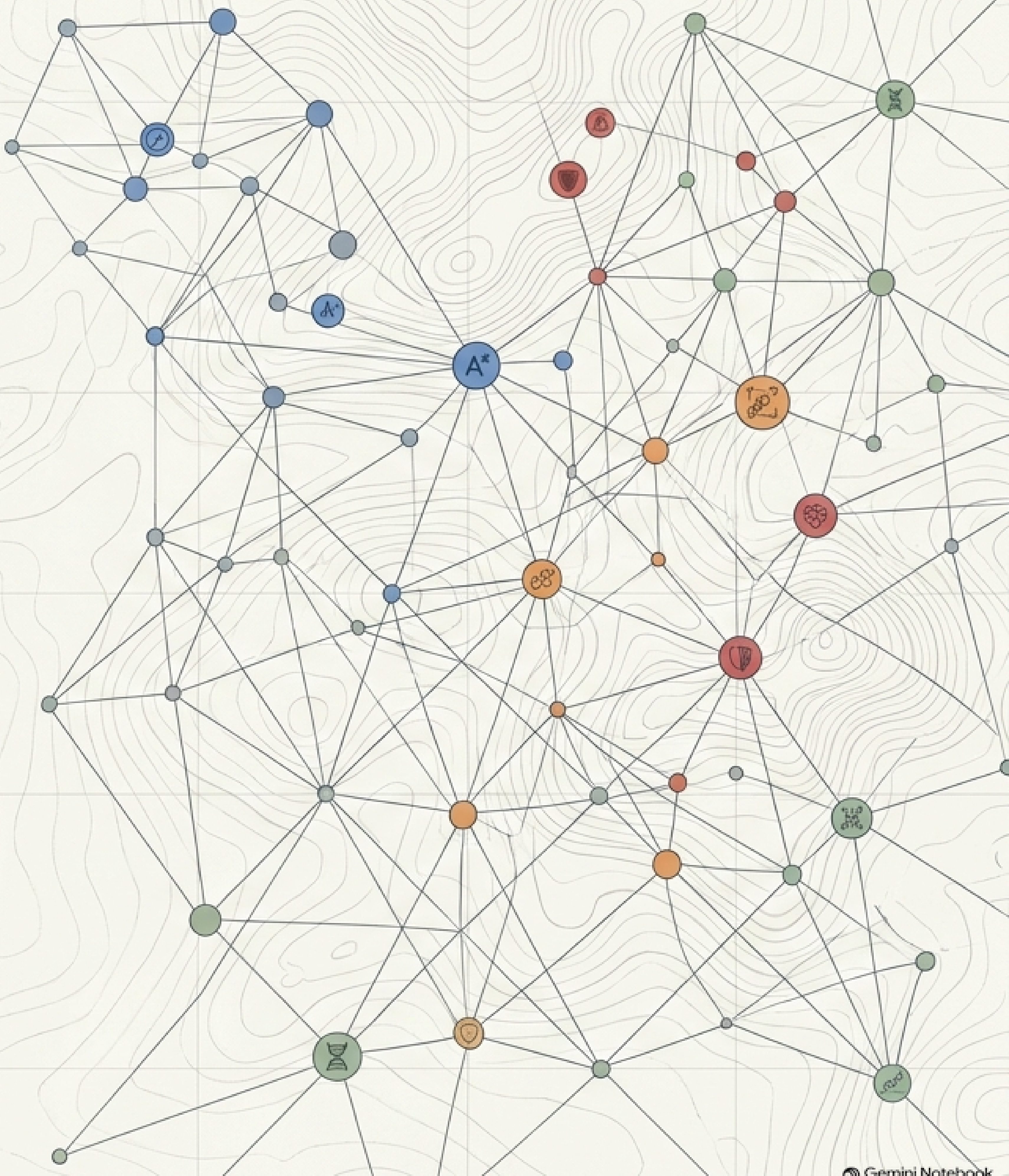
# Navigating the Impossible: AI Heuristics and Problem-Solving Techniques

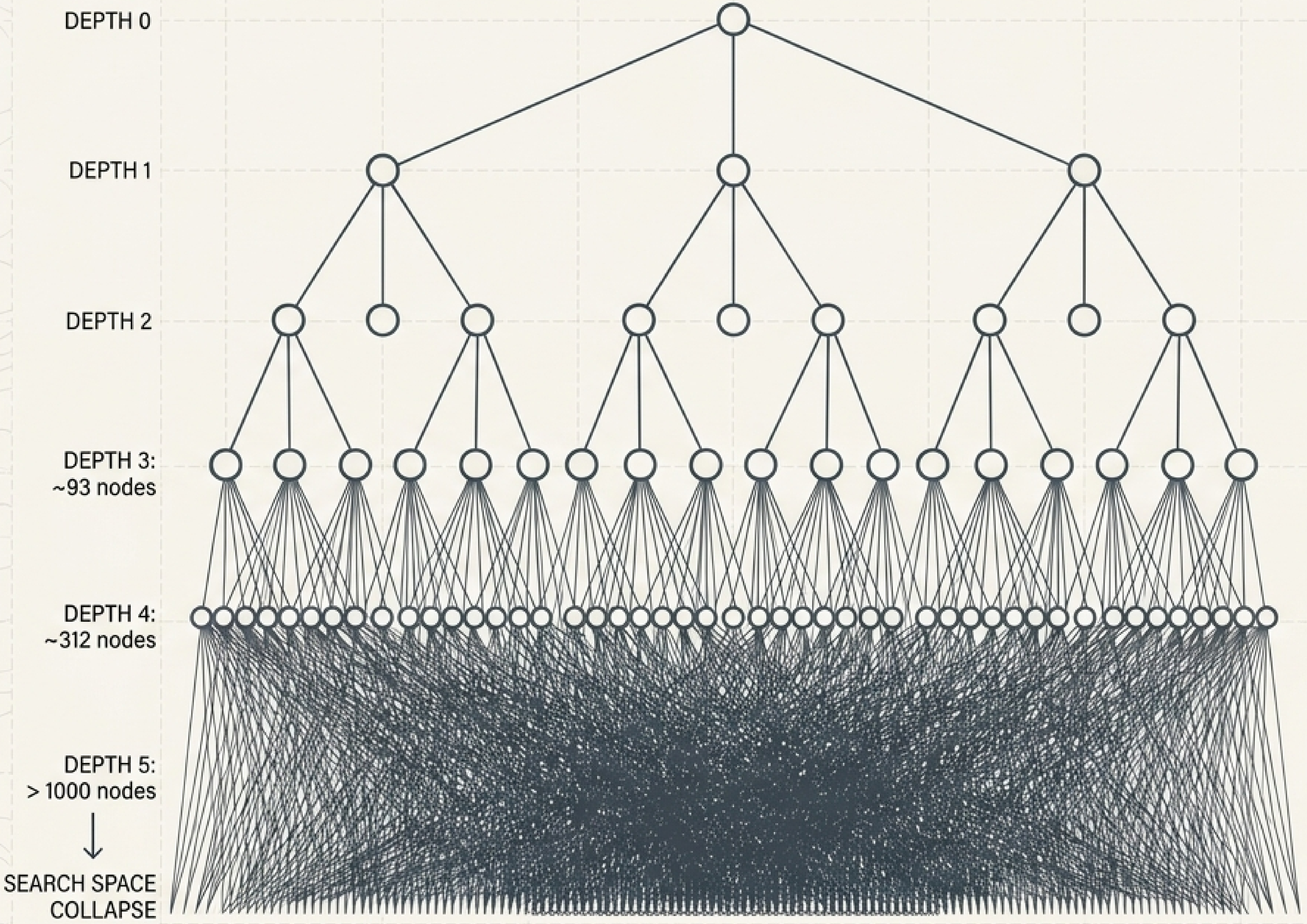
A Strategic Framework for Search, Optimization, and Adversarial Decision-Making



## The Core Problem: Combinatorial Explosion

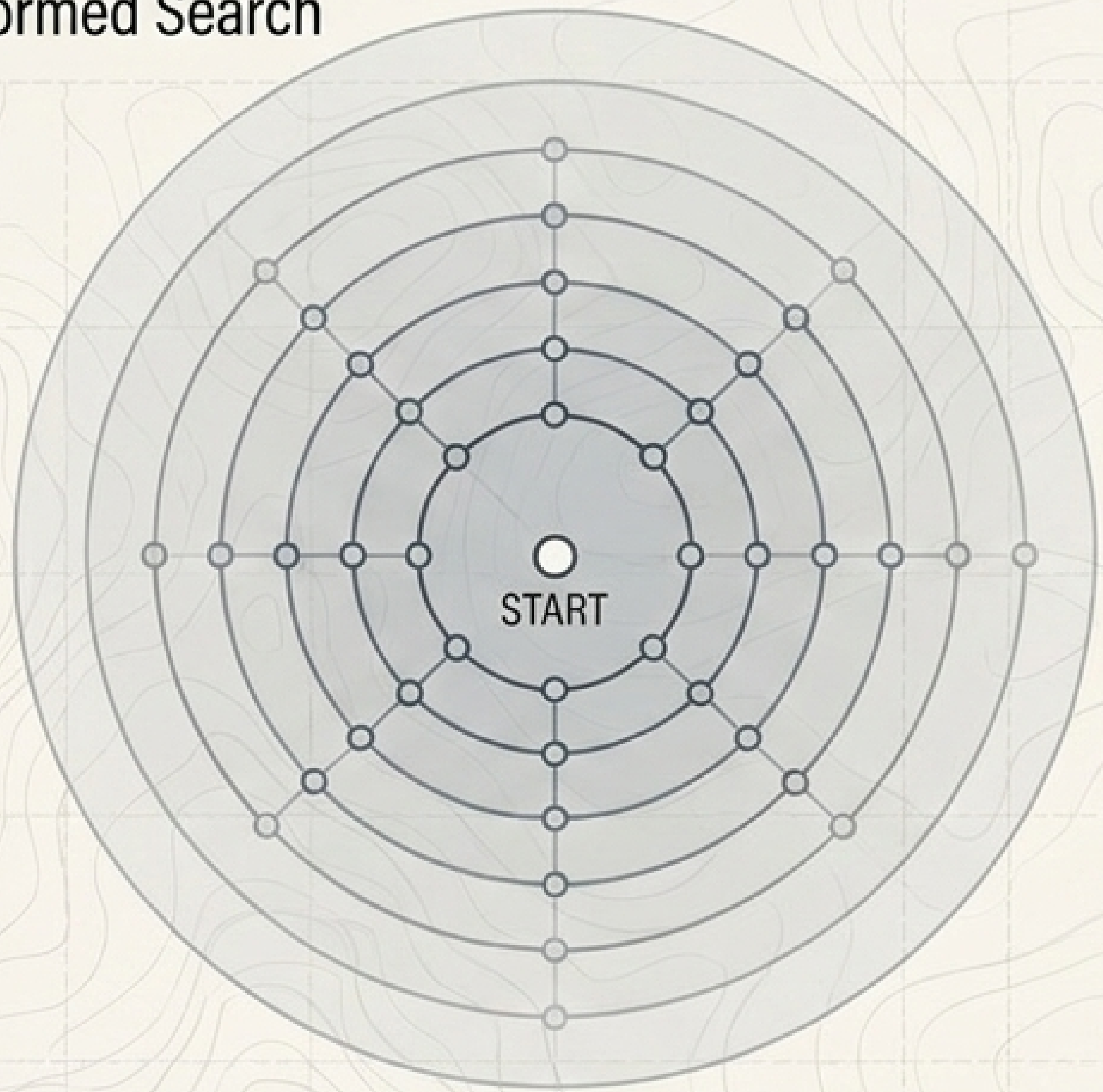
$$\sum_{i=0}^{d} b^i = \frac{b^{d+1} - 1}{b - 1}$$

A problem with b possible actions at each state and a solution depth of d grows exponentially. Blind search fails. Intelligent search techniques are mandatory to survive the math.

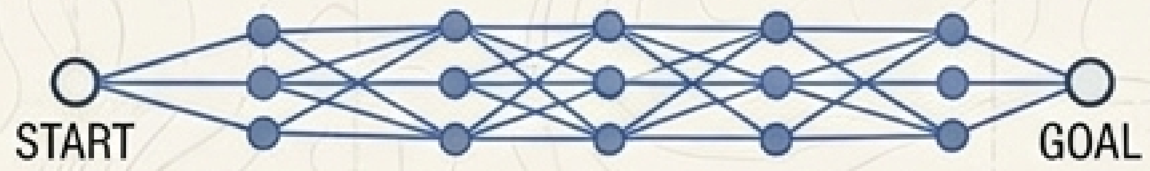


# The Antidote: Heuristic Search

## Uninformed Search




## Informed Search

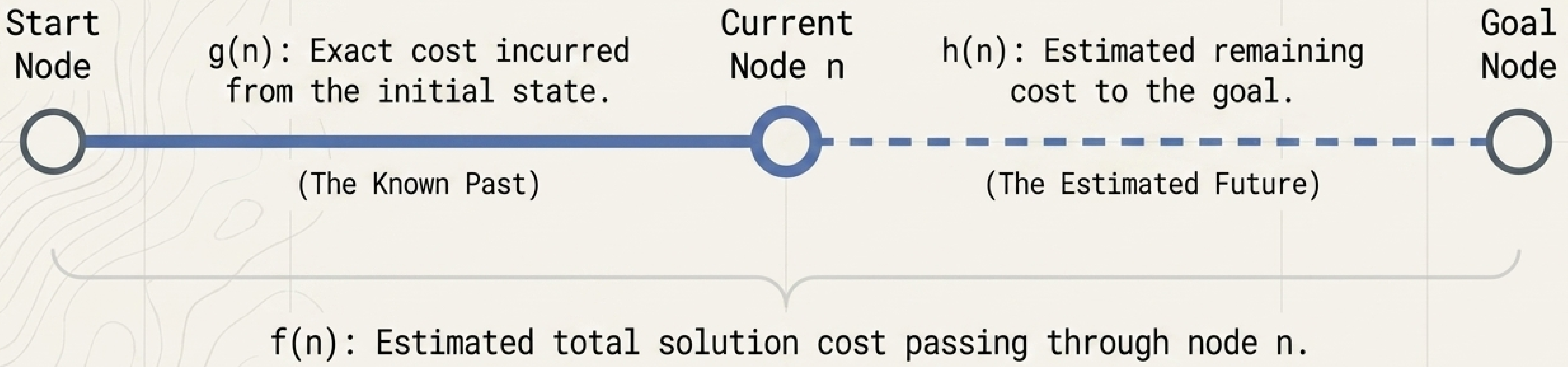



**Definition:** A heuristic, $h(n)$, is a rule or estimate using problem-specific domain knowledge to guide a search toward promising solutions.

**The Principle:** Intelligence is knowing what not to compute. Prioritize states that mathematically appear closer to the goal.

# A* Search: Balancing the Past and the Future

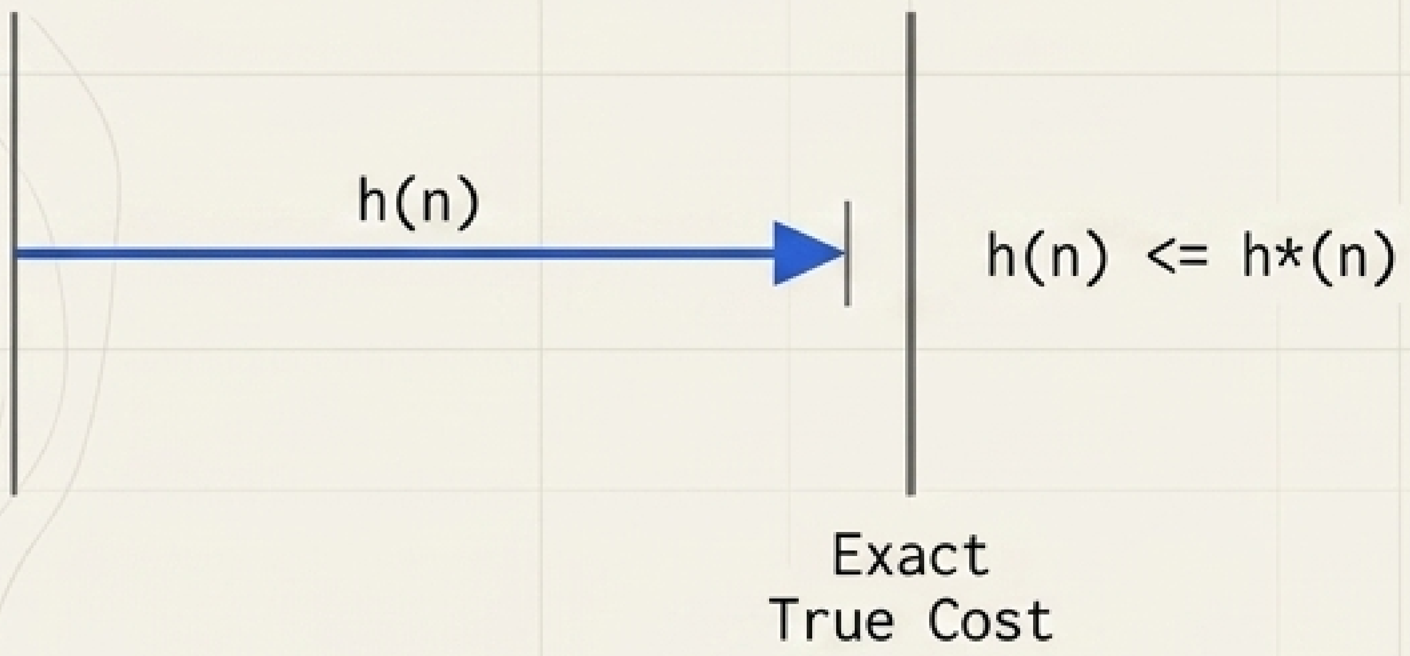
$$f(n) = g(n) + h(n)$$




**Takeaway: Unlike Greedy Search which only looks at h(n), A* expands the node with the lowest total f(n), ensuring optimal pathfinding.**

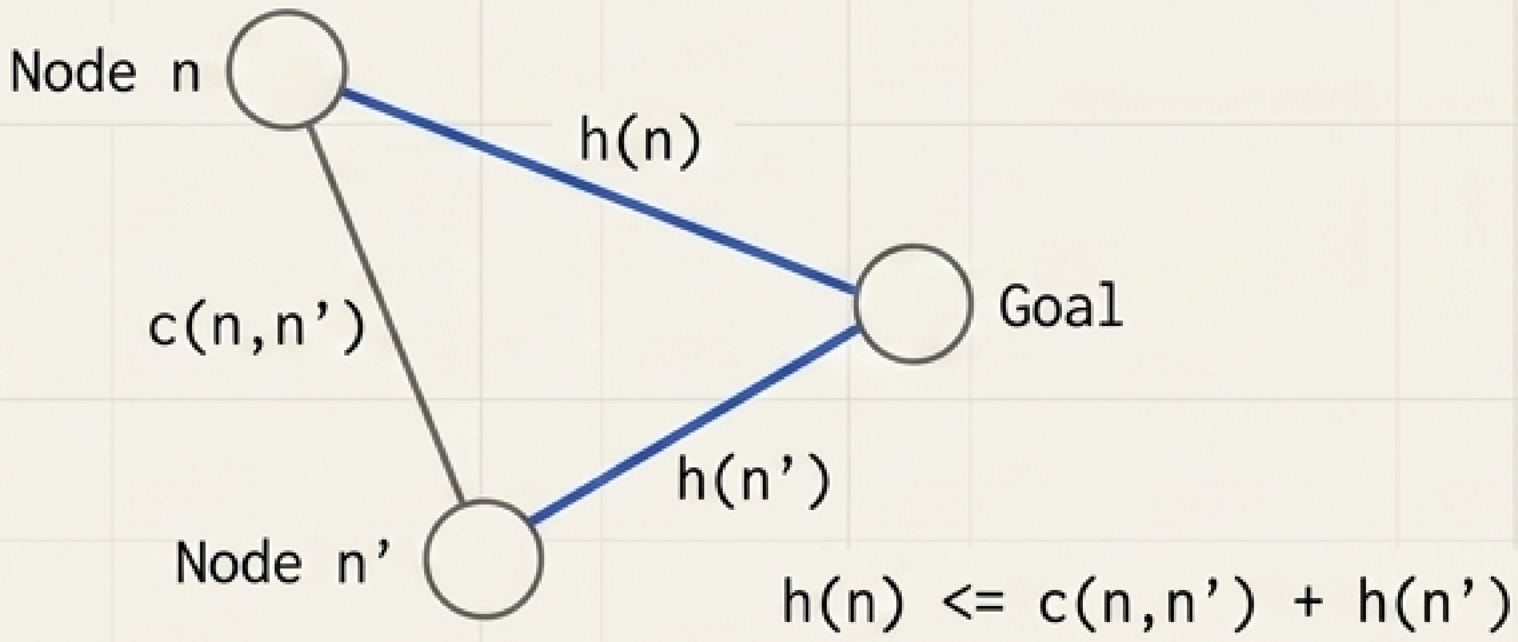
# The Rules of the Estimate: Guaranteeing Optimality

## Admissible (Optimistic)



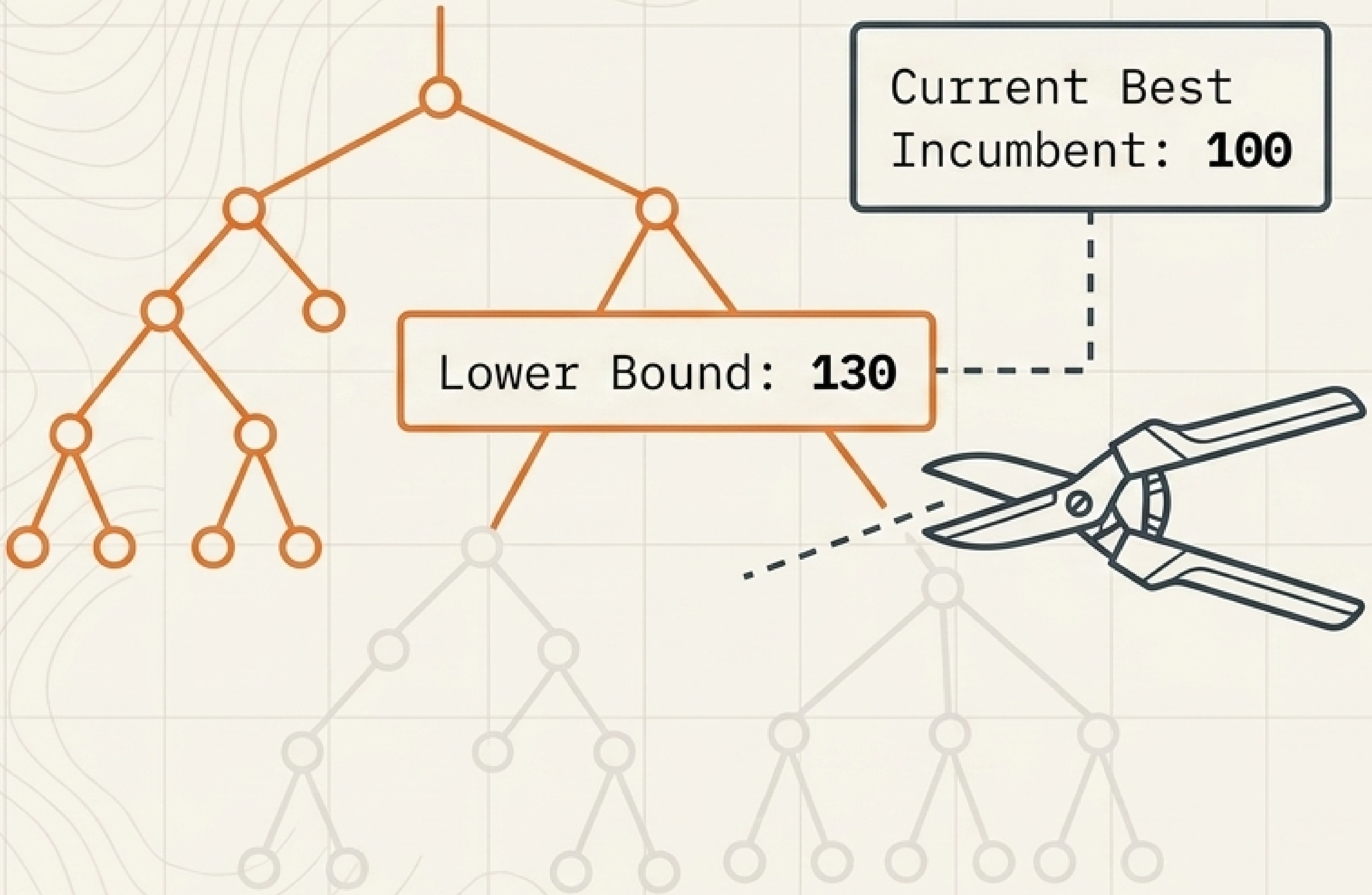

The heuristic must never overestimate the true minimum cost. It provides a mathematical lower bound.

## Consistent (Logical)




The estimated cost cannot decrease by more than the actual transition cost between steps.

# Branch & Bound: Exact Optimization via Pruning




## The Process

1. Branch the subproblems.
2. Compute a lower bound.
3. Prune realities that cannot mathematically beat the incumbent solution.

## Core Difference

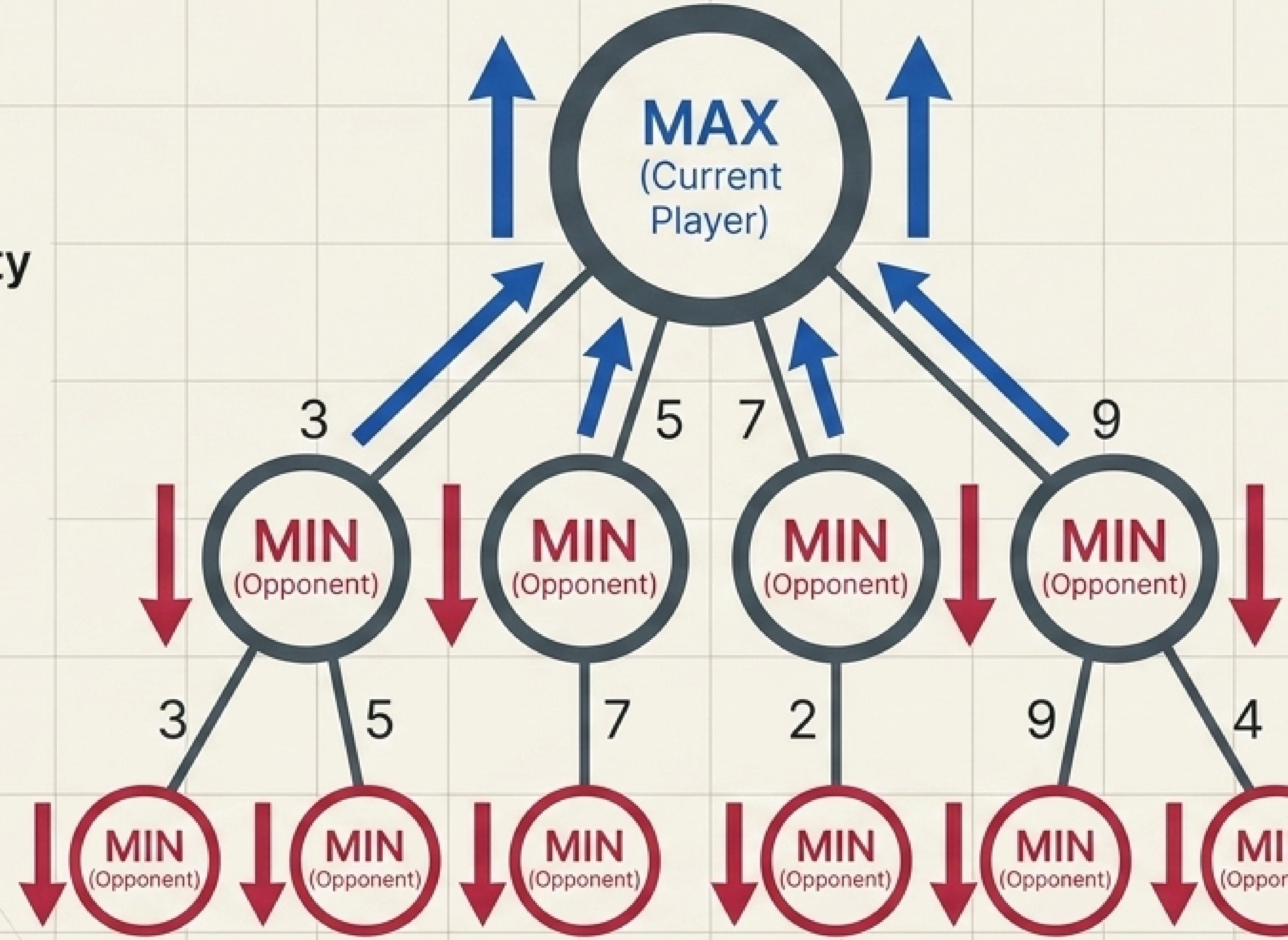
A* is for goal/path search; Branch & Bound is explicitly for optimizing and maintaining an incumbent best solution.

# Game Theory & Minimax: Adversarial Search

**The Zero-Sum Reality**

$U_A + U_D = 0$

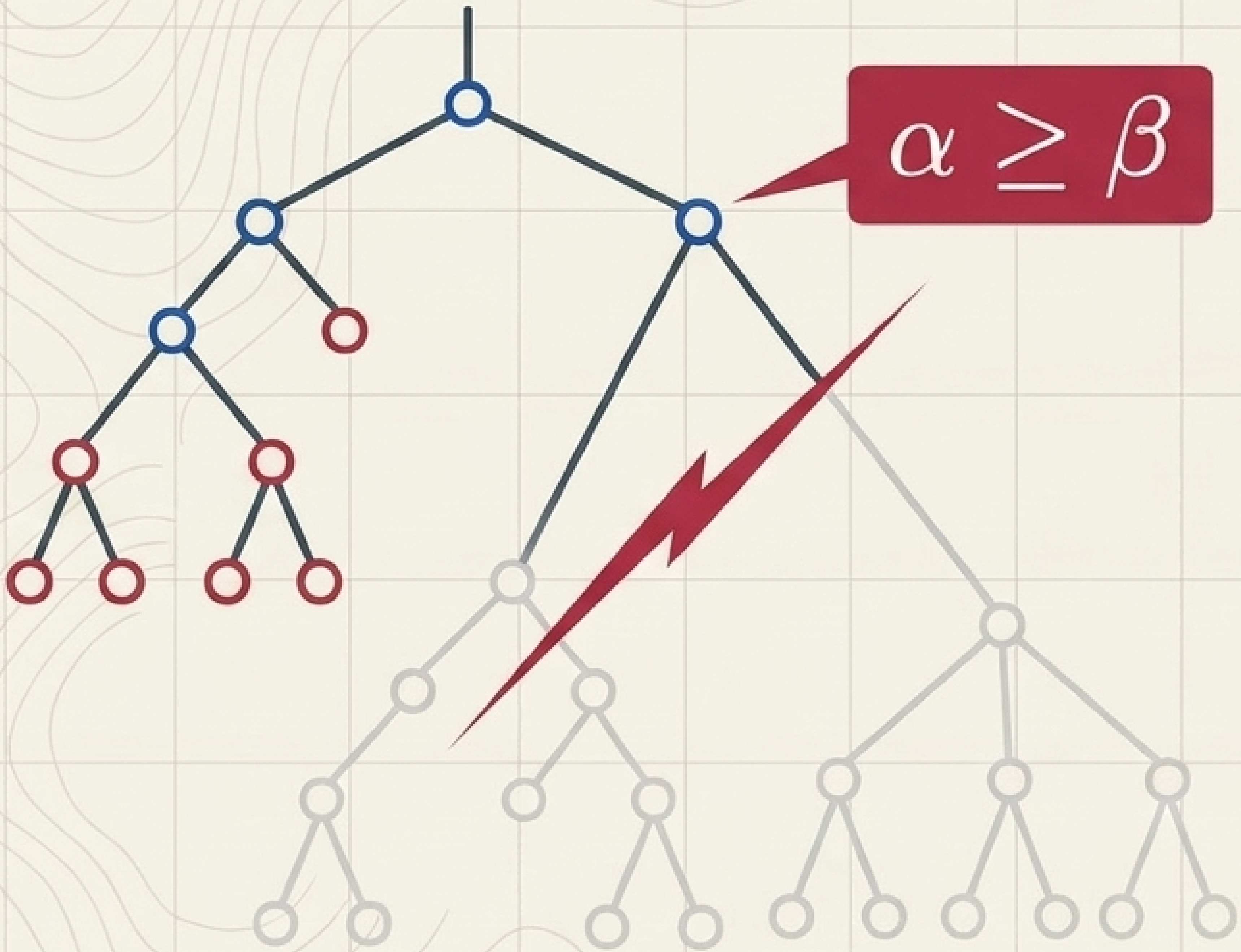
One player's gain is the other's exact loss.




**The Minimax Principle**

Assume the opponent will choose the action that is worst for you. Optimize your strategy against that worst-case reality (max vs min).

# Alpha-Beta Pruning: Accelerating the Game




**Mechanism**

Tracks $\alpha$ (best value guaranteed for MAX) and $\beta$ (best guaranteed for MIN). If $\alpha \geq \beta$, evaluate no further.

**The Impact**

Reduces search complexity from a naive $O(b^d)$ down to $b^d$ under optimal move ordering. Do not evaluate what cannot change the final decision.

# Evolutionary Algorithms: Biological Optimization

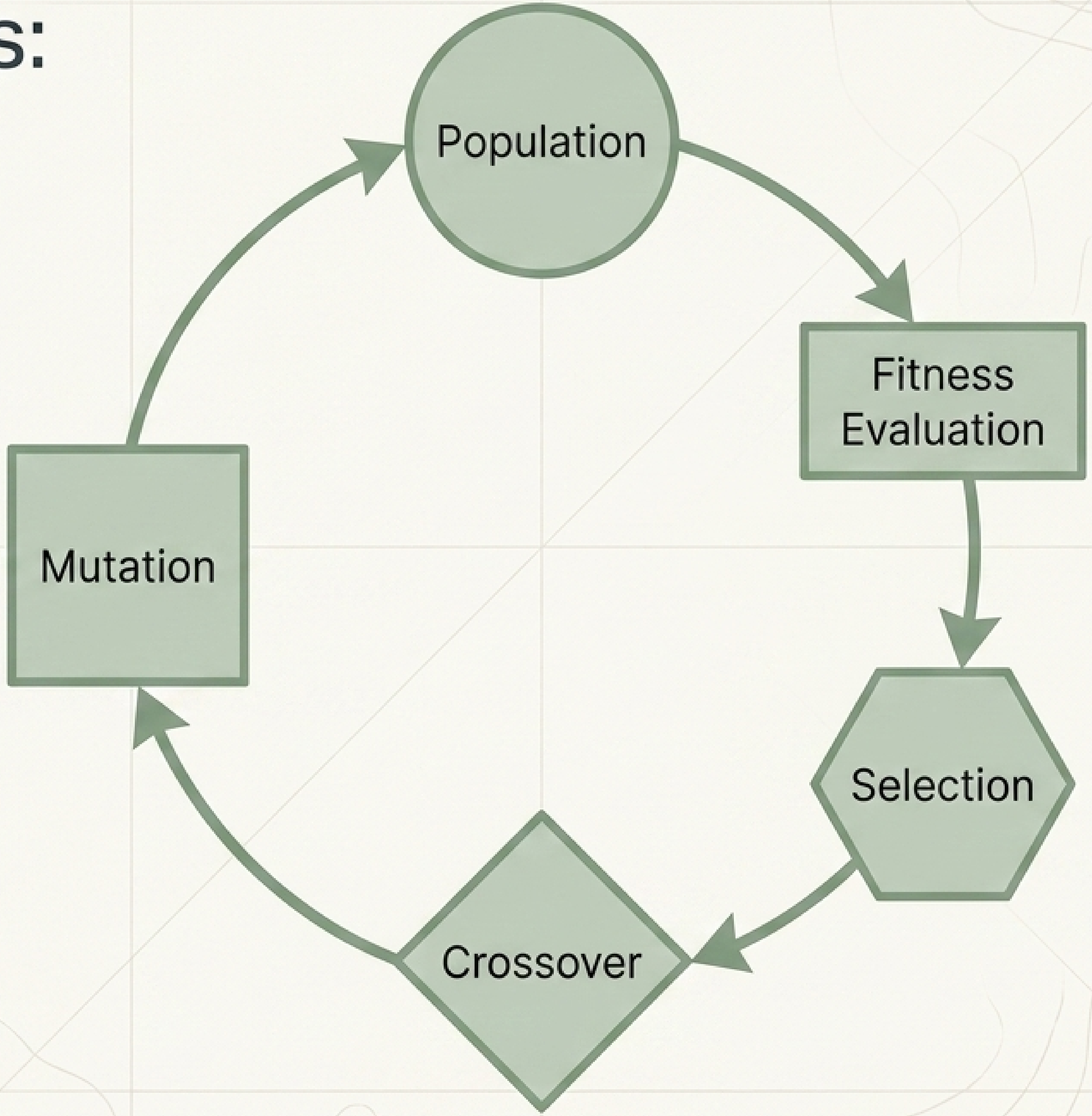
**Motivation**

Deployed when search spaces are too vast, discrete, or non-linear for exact math.

---

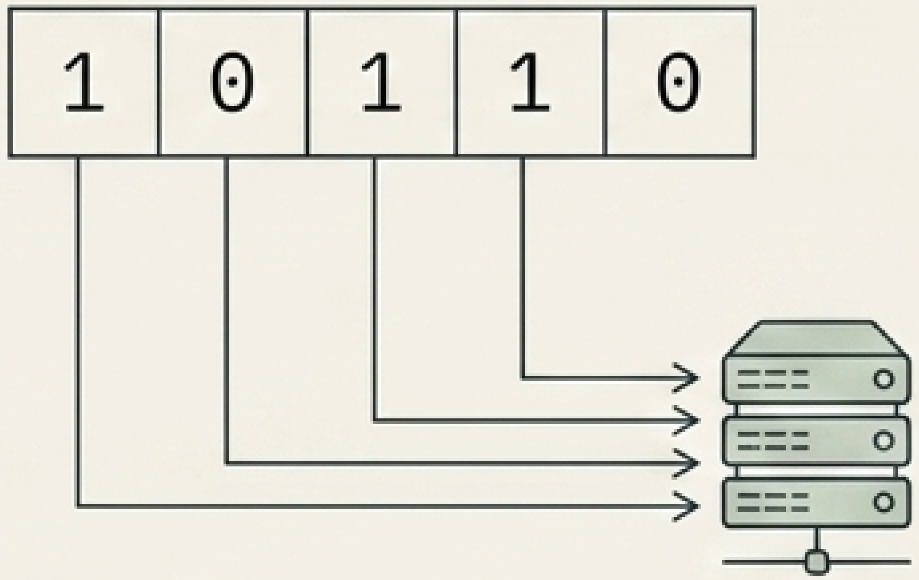
**The Core Rule**

An optimization algorithm only optimizes what the objective function measures. Fitness function design is critical.



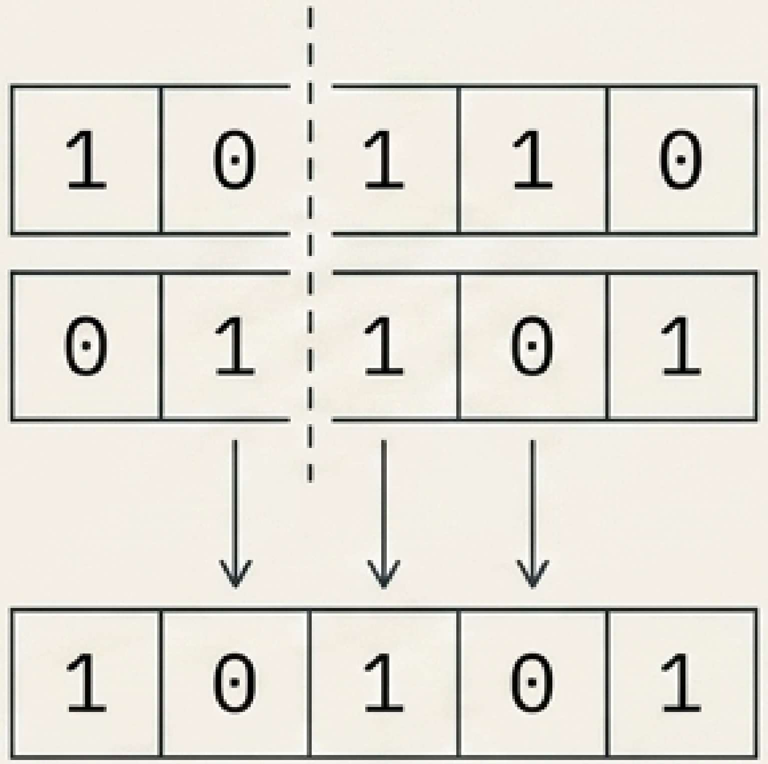
# The Mechanics of Genetics in AI

## 1 Representation



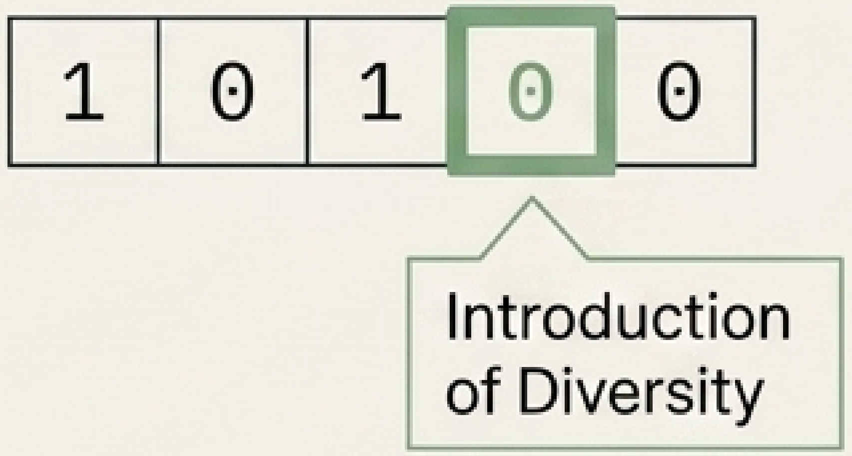

Encoding solutions (e.g., binary chromosomes or continuous vectors).

## 2 Crossover



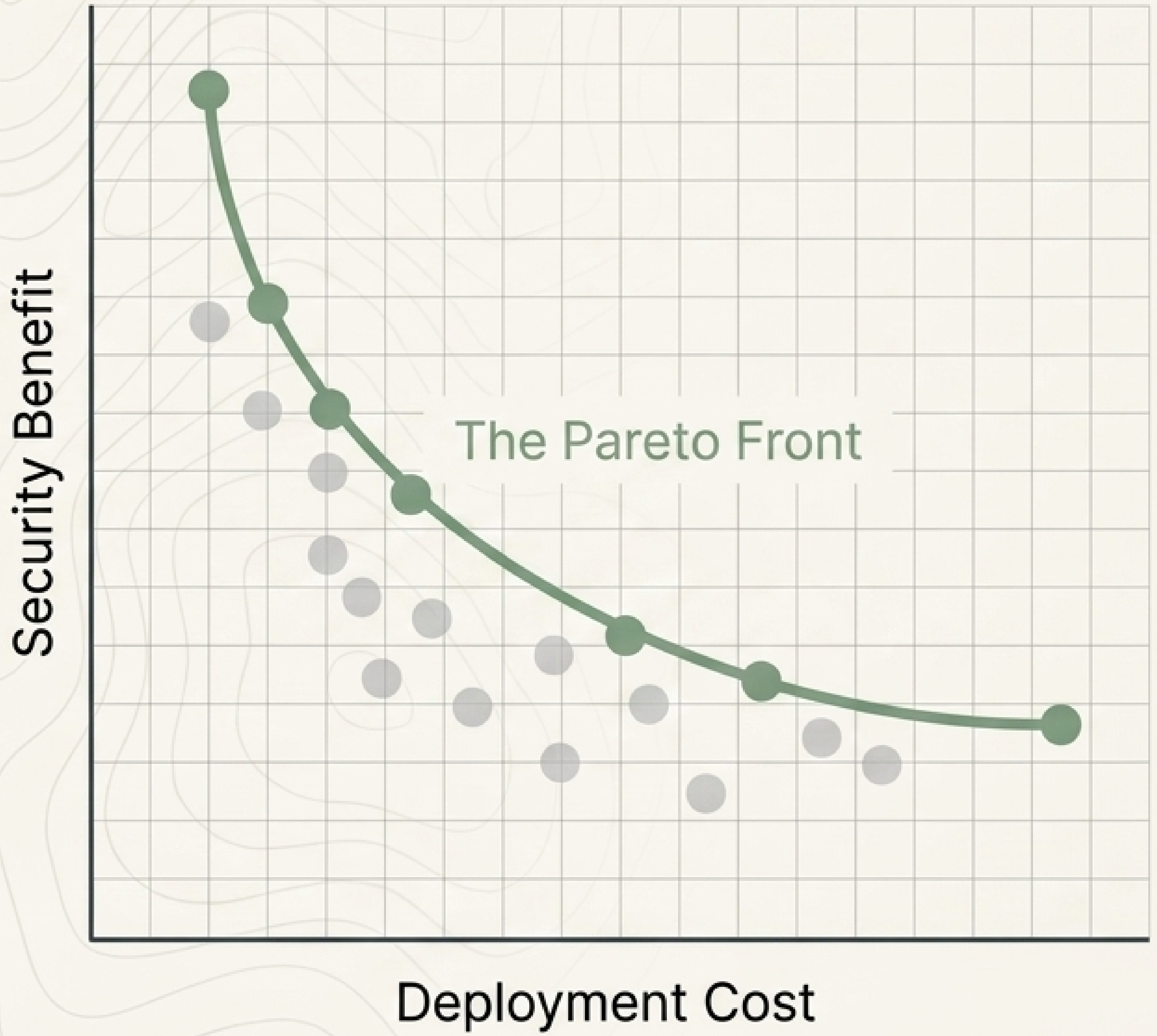

Recombining successful traits from multiple parents.

## 3 Mutation




Introducing random variation to prevent premature convergence on local optima.

# Multi-Objective Optimization & The Pareto Front




### The Equation

`Fitness(x) = Throughput(x) - λCost(x)`

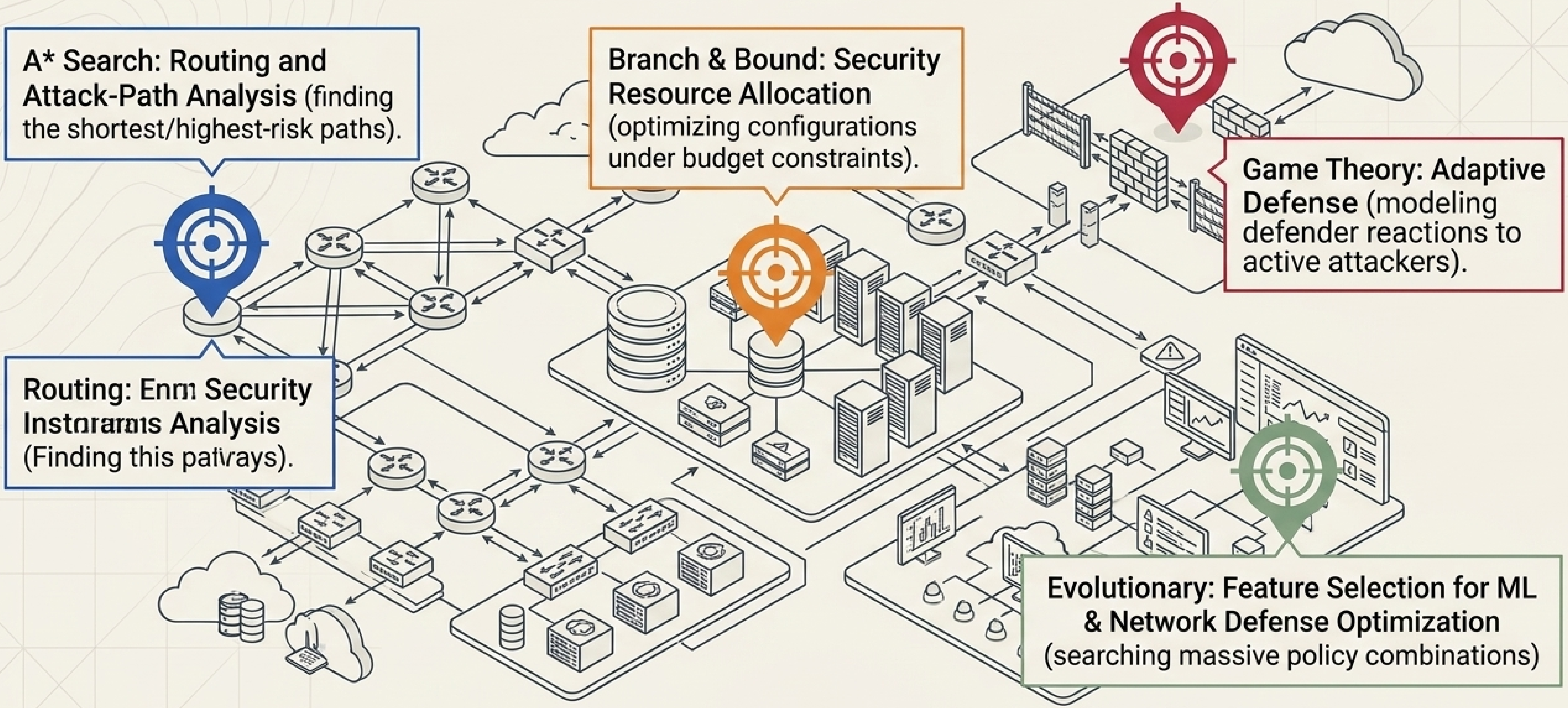
### Pareto Dominance

A solution dominates if it is at least as good in every objective and strictly better in one.

### The Result

Evolutionary methods return a frontier of optimal compromises, leaving the final operational choice to the human engineer.

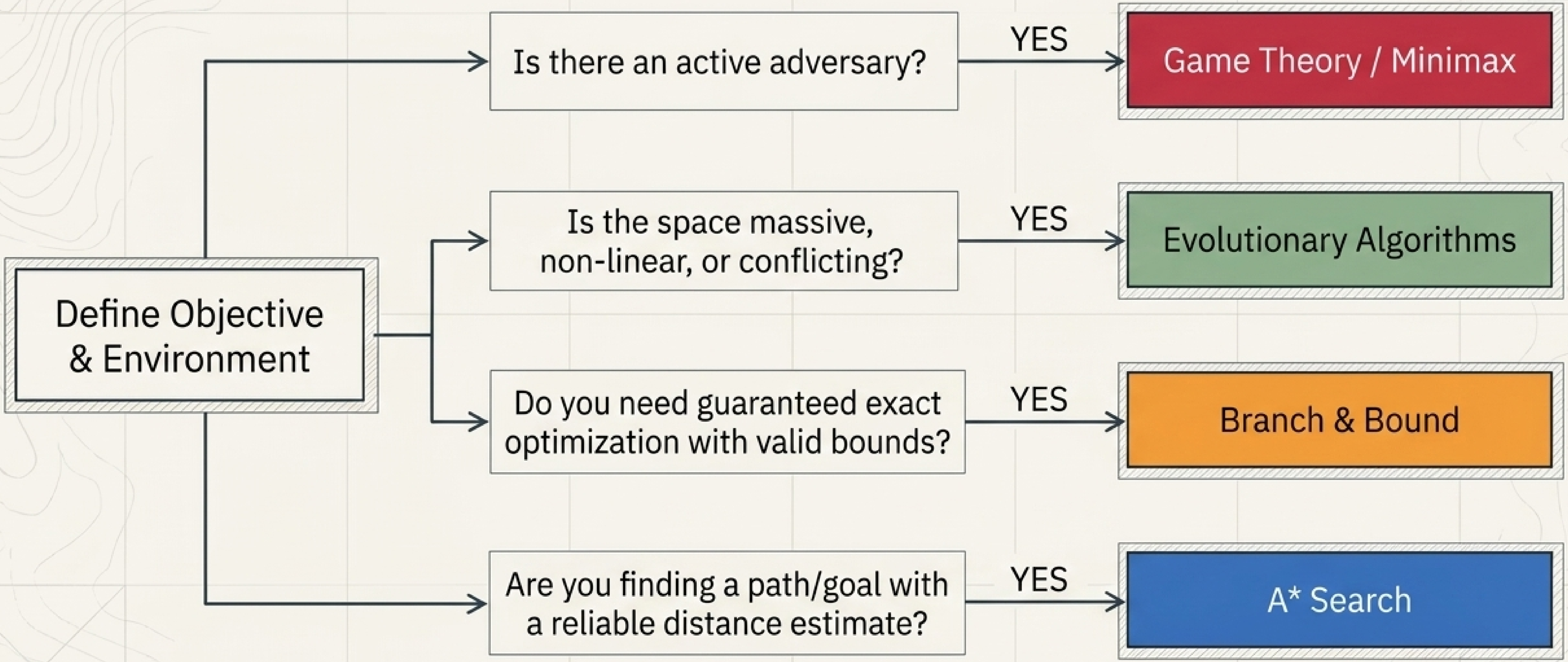
# Domain Application: Securing the Network



# The Diagnostic Matrix: Comparing Techniques

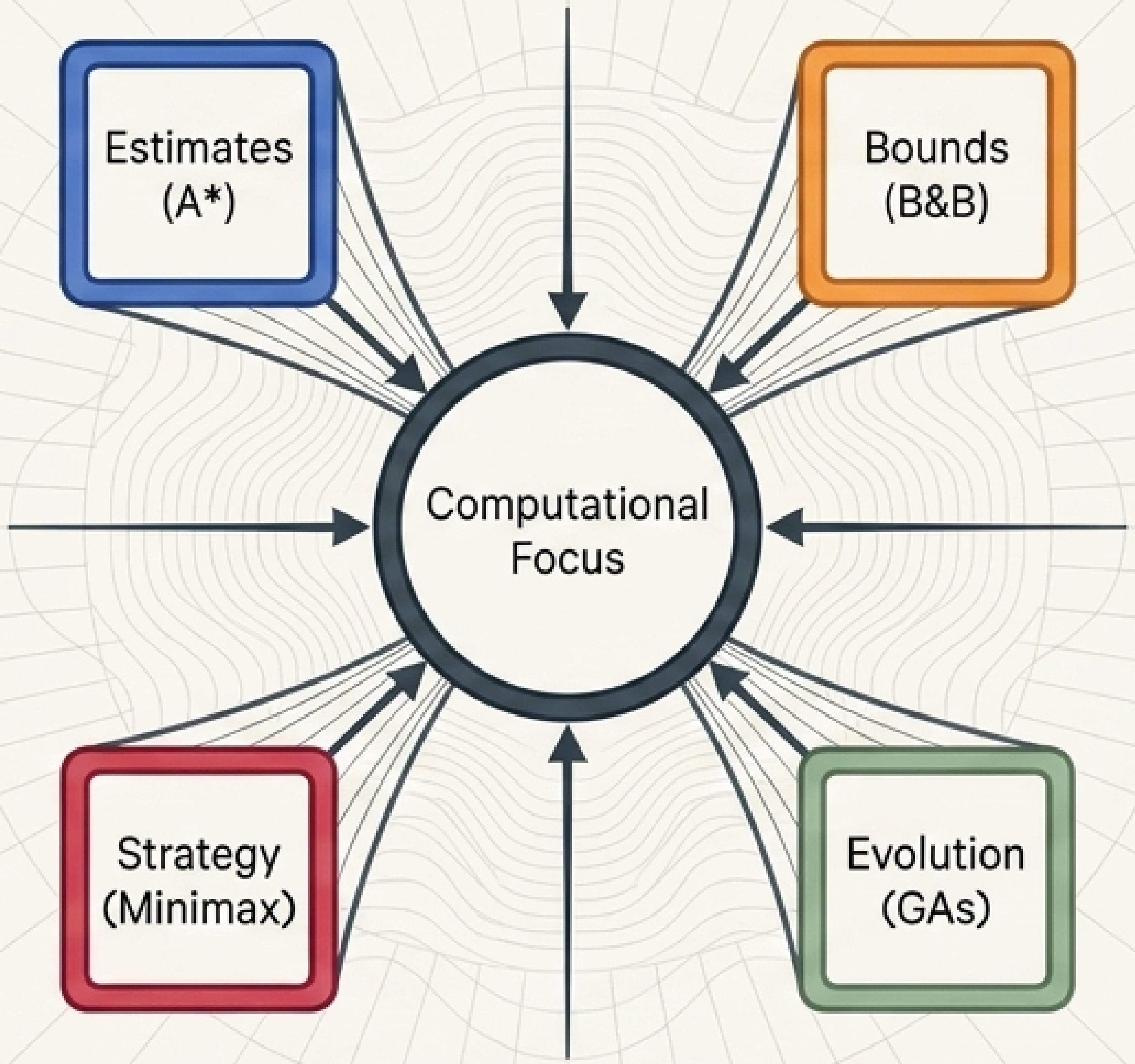
| Technique | Core Mechanism | Exact or Approx? | Primary Limitation |
|---|---|---|---|
| A* Search | $g+h$ guided search | Exact (with admissible heuristic) | Depends heavily on heuristic quality. |
| Branch & Bound | Branch + bound + prune | Exact optimization | Can still be exponentially slow. |
| Minimax/Alpha-Beta | Assume adversarial opponent | Exact strategic decision | Bound by depth and game tree size. |
| Genetic Algorithms | Evolve candidate populations | Approximate / Flexible | No guarantee of global optimality. |

# The Decision Framework: How to Choose




**Diagnostic Note:** Always estimate required guarantees: Exact optimum vs. ‘Good enough, fast’.

# The Unified View




**THE CENTRAL LESSON: DO NOT BLINDLY COMPUTE EVERY POSSIBILITY.**

Whether using estimates, lower bounds, strategic opponent reasoning, or biological feedback, intelligence is simply the use of structure to focus limited effort on promising realities.